UNIT I | SLOT C CSA1603 - DATA WAREHOUSING & DATA MINING

OLAP and Data Cubes

Online Analytical Processing — multidimensional analysis for fast, flexible decision support

Presented by **Trisha N 192311398**

Data Warehousing and Data Mining — Unit I

ROADMAP

What We'll Cover

CONCEPT

What is OLAP?

OLAP (Online Analytical Processing) is a technology that enables users to interactively analyze multidimensional data from multiple perspectives — quickly, and directly.

1. **Fast**

   Sub-second response even on huge, aggregated datasets

2. **Analysis**

   Supports complex calculations, trend and what-if analysis

3. **Shared**

   Multiple users can query and analyze concurrently

4. **Multidimensional**

   Data viewed across dimensions like time, product, region

THE FASMI TEST

Codd's 5 defining rules for OLAP systems

- **F** Fast — response within seconds
- **A** Analysis — rich calculations built in
- **S** Shared — secure, multi-user access
- **M** Multidimensional — data cube views
- **I** Information — access all relevant data

COMPARISON

OLAP vs OLTP

Decision support vs day-to-day transaction handling

Aspect	OLTP (Operational)	OLAP (Decision Support)
Purpose	Run day-to-day operations	Analyze data for decisions
Data	Current, detailed, up-to-date	Historical, summarized, consolidated
Operations	Insert, update, delete	Read-heavy, complex queries
Design	Normalized (ER model)	Denormalized (star/snowflake)
Users	Clerks, cashiers, DB admins	Managers, analysts, executives
Response time	Milliseconds	Seconds to minutes

ARCHITECTURES

Types of OLAP Systems

MOLAP

Multidimensional OLAP

- Data stored in proprietary cube format
- Very fast retrieval & pre-computed aggregates
- Higher storage overhead
- Best for smaller, dense datasets

ROLAP

Relational OLAP

- Data stays in relational tables (RDBMS)
- Uses SQL to compute aggregates on the fly
- Scales to very large datasets
- Slower than MOLAP for complex queries

HOLAP

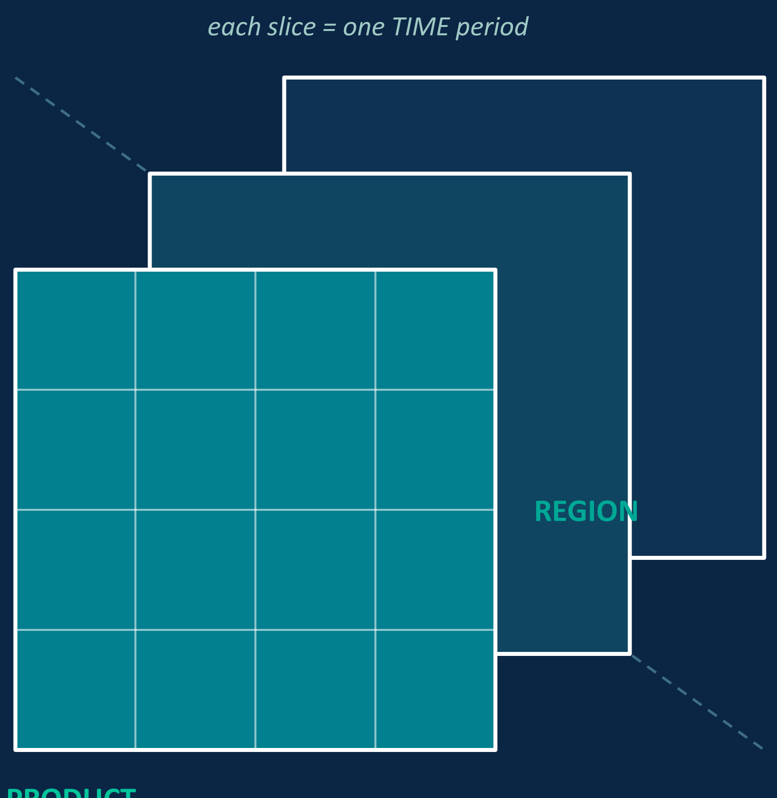
Hybrid OLAP

- Combines MOLAP speed with ROLAP scale
- Summary data in cubes, detail in relational
- Balanced storage and performance
- Flexible for mixed workloads

THE MULTIDIMENSIONAL MODEL

The Data Cube

A data cube organizes facts (measures) along multiple dimensions, letting analysts view figures like sales from any angle — by time, product or location.



Key Terms

Dimension:

A perspective for analysis — e.g. Time, Product, Region, Customer

Measure:

The numeric fact being analyzed — e.g. Sales, Revenue, Quantity

Cell:

One intersection of all dimensions — a single aggregated value

Cuboid:

A cube computed at one level of aggregation (group-by)

Lattice of Cuboids:

All possible cuboids from base cuboid to apex cuboid

NAVIGATION

OLAP Operations

How analysts navigate a data cube to explore it from different angles

1	**Roll-up**	Aggregates data by climbing up a concept hierarchy (e.g. city -> state -> country)
2	**Drill-down**	Reverses roll-up — moves from summarized to more detailed data
3	**Slice**	Selects one dimension value, producing a single sub-cube (a 2D view)
4	**Dice**	Selects a sub-cube by choosing specific values on two or more dimensions
5	**Pivot (Rotate)**	Rotates the cube's axes to view data from a different orientation

IMPACT

Why OLAP Matters

Advantages

- Enables fast, interactive analysis of huge datasets
- Supports complex, multidimensional what-if calculations
- Improves decision-making with historical trend views
- Reduces load on operational (OLTP) systems
- Presents data intuitively for non-technical users

Real-World Applications

- Sales & marketing performance dashboards
- Financial reporting and budget forecasting
- Retail and inventory demand analysis
- Business intelligence tools (KNIME, Pentaho)
- Customer segmentation and churn analysis

Key Takeaways

- OLAP enables fast, multidimensional analysis for decision support (FASMI).
- It differs from OLTP by favoring read-heavy, historical, aggregated analysis.
- MOLAP, ROLAP and HOLAP trade off speed, storage and scalability differently.
- Data cubes organize measures across dimensions, computed as a lattice of cuboids.
- Roll-up, drill-down, slice, dice and pivot let analysts explore the cube interactively.

Thank You